

### 7.3. Differentiable dynamics - generic properties.

Def. A property  $P(x)$  in  $X^r(M)$  is **generic**  
if  $\{x \in X^r(M) : P(x)\}$  is residual.

A property  $P(x)$  in  $X^r(M)$  is  **$C^r$  generic**  
if  $\{x \in X^r(M) : P(x)\} = X(M) \cap \text{subspace of } X^r(M)$   
containing residual set.

Rmk.  $X^r(M) : C^r$  generic in  $X(M)$ .

•  $X(M) \subset X^r(M) : \text{residual}$   
•  $X^r(M) : \text{Baire}$ .  $\Rightarrow X(M) : \text{dense in } X^r(M)$

i.e.  $P(x) : \text{generic} \Rightarrow$  every  $x$  can be approximated  
as we wish by one with the generic property.

Def.  $X \in X(M)$  has property **(G2)**

if every critical element is hyperbolic

$\lambda, \alpha_1, \dots, \alpha_m \quad |\alpha_i| \neq 1 \quad \forall i.$

Rmk. (G2)  $\Rightarrow$  Crit. pts are isolated

& have stable & unstable mds.

$S^+(\gamma)$

$S^-(\gamma)$

Def.  $X \in X(M)$  has property **(G3)**

if  $X$  has (G2) &  $m \in S^+(\gamma) \cap S^-(\delta)$  for any  $\gamma, \delta$

Thm. (Kupka-Smale)  $\Rightarrow T_m S^+(\gamma) \nparallel T_m S^-(\delta)$

$M : \text{cpt}, r \geq 1 \Rightarrow (G3) : C^r \text{ generic.}$

## 8.4. Hamiltonian dynamics - generic properties.

\*  $\mathcal{H}$ -elementary.

- Zero : Not a char. exponent
- $\lambda$  : char. exponent w/  $\operatorname{Re} \lambda = 0 \rightarrow$  multiplicity 1.
- $\lambda, \mu$  : char. exponents,  $\operatorname{Re} \lambda, \operatorname{Re} \mu = 0$ ,  $\operatorname{Im} \lambda, \operatorname{Im} \mu > 0$   
 $\Rightarrow \lambda \& \mu$  : indep /  $\mathbb{Z}$ .

Def.  $H \in \mathcal{F}(M)$  or  $X_H \in \mathcal{X}_H(M)$  has (H1)

if every crit pt is  $\mathcal{H}$ -elementary.

Thm. (Burchner).

(H1) :  $C^k$  generic in  $\mathcal{X}_H(M)$   $\forall k \geq 1$ .

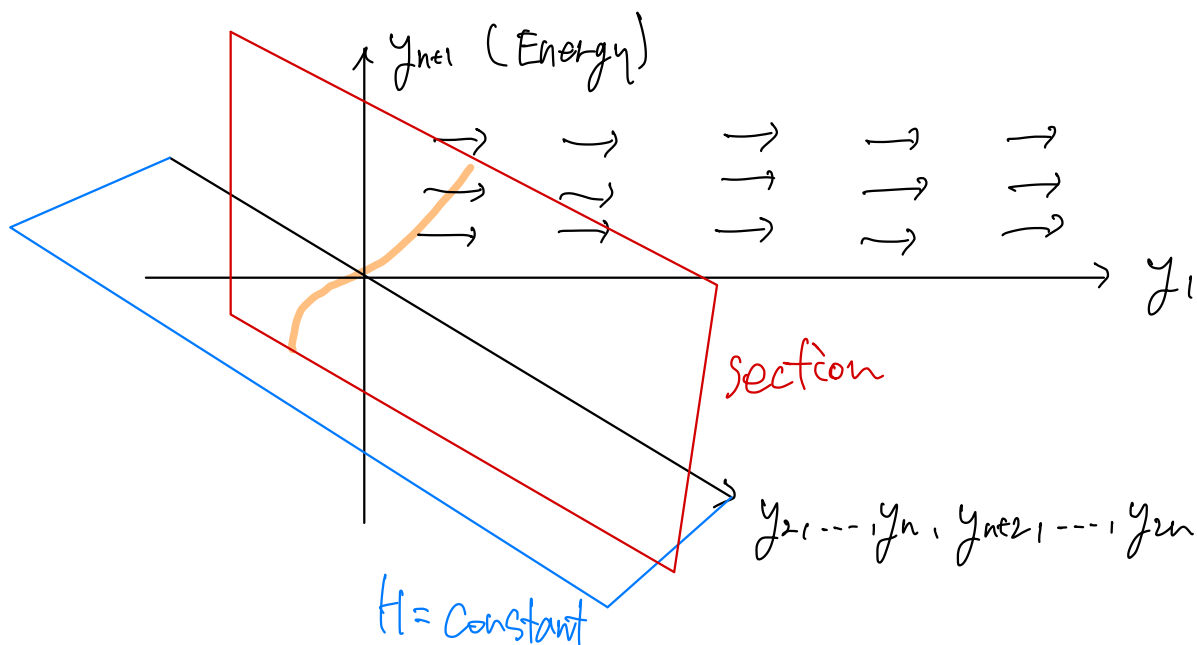
$D\alpha \circ \theta - D\alpha$  : surj.

\*  $\gamma$  : 0-elementary closed orbit of  $X_H$ .

$m \in \gamma$ ,  $(U, \phi)$  : power chart at  $m$ .

$$(y_1, \dots, y_n) \quad H(y) = y_{n+1}$$

$$\dot{y}_1 = 1, \quad \dot{y}_2 = \dots = \dot{y}_n = 0.$$



\*  $\gamma$ : 0-elementary

$(u, \phi)$ : power chart at  $m$ .

$$\phi: U \rightarrow \mathbb{R} \times \mathbb{R}^{n-1} \times \mathbb{R} \times \mathbb{R}^{n-1}$$

$$u \mapsto (\delta, q, \varepsilon, p)$$

$$\Theta': W'_0 \longrightarrow W'_1 \subset \mathbb{R}^{n-1} \times \mathbb{R} \times \mathbb{R}^{n-1}$$

$$(q, \varepsilon, p) \longmapsto (Q, \varepsilon, P)$$

: local representation of Poincaré map.

transversal section: defined by  $\delta = 0$ .

$$\Theta_\varepsilon: W'_{00} \longrightarrow W'_{1,\varepsilon} \subset \mathbb{R}^{n-1} \times \mathbb{R}^{n-1}$$

$$(q, p) \longmapsto (Q(q, \varepsilon, p), P(q, \varepsilon, p))$$

$$\text{s.t. } W'_{00} \times (-a, a) \subset W'_0$$

$$\varepsilon \in (-a, a) \Rightarrow \phi_\varepsilon: W'_{00} \rightarrow W'_{1,\varepsilon}: \text{symp. diffeo.}$$

$\gamma_\varepsilon$  of orbit cylinder: represented by fixed pt  $(q_\varepsilon, p_\varepsilon)$  of  $\phi_\varepsilon$ .

$$D_\varepsilon(q_\varepsilon, p_\varepsilon) \in \text{Sp}(\mathbb{R}^{n-1} \times \mathbb{R}^{n-1})$$

$\downarrow$

$$D_\phi \Theta_\varepsilon: (-a, a) \rightarrow \text{Sp}(\mathbb{R}^{2n-2}).$$

tangent arc of  $\gamma_0$  w.r.t. the power chart  $(u, \phi)$ .

Def.  $X_H$  has property (H2) if

- $\forall$  closed orbits are 0-elementary.

$$\forall N \in \mathbb{Z}_+,$$

- $\{\gamma_e : e \in (-a, a)\} : \text{orbit cyl.}$

$$D\phi|_{\Theta_e} : (-a, a) \rightarrow Sp(\mathbb{R}^{2n-2})$$

tangent arc w.r.t. p.c.

transverse frequencies  $\{\alpha_i\}$  are

lin. indep. /  $\{-N, \dots, N\}$

for all but finitely many  $e \in (-a, a)$ .

Thm (Robinson)

$$(H2) : C^t \text{ generic } \forall t \geq 2.$$

(Sketch) • 0-elementary cond: transversality

$$B_N := \left\{ A \in Sp(\mathbb{R}^{2n-2}) : \begin{array}{l} \exists p = (p_i) \in \{-N, \dots, N\} \\ \text{s.t. resonance cond.} \end{array} \right\}$$

$\cup$   
codim 1 hypersurface.

- tangent arc  $\cap B_N : \text{generic}$

$\Rightarrow$  Only finitely many resonance

- $\cap$  over  $N \in \mathbb{Z}_+$ .

Def.  $X_H$  has property (H3) if

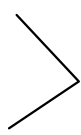
- $X_H$  has (H2)

- $\forall$  equilibrium pts lie on "different" energy surfaces.

- $m$  : equilibrium pt of  $X_H \Rightarrow W^+(m) \cap W^-(m)$

- $m$  : equilibrium

- $\gamma$  : closed orbit



in the same energy surface

in  
condz.

$$\Rightarrow W^+(m) \cap W^-(\gamma) \text{ \& } W^-(m) \cap W^+(\gamma)$$

- $T, \Delta$  : regular orbit cyl  $\Rightarrow W^+(T) \cap W^-(\Delta)$

- For all but countable set  $B$  of closed orbits,

$$W^+(\gamma) \cap W^-(\delta)$$

- $\forall$  bad closed orbit  $\beta \in B$  is interior to a regular orbit cyl  
( $\beta$  is not PCM)

Thm (Robinson)

(H3) is  $C^1$  generic  $\forall r \geq 2$ .

Summary.

(G2) : crit pts : hyperbolc.

(G3) : (G2) & stable & unstable

(H2) : non-degeneracy of closed orbits

& finitely many resonance in orbit cyl.

(H3) : (H2) & transversality in energy surface.

(except countably many bad orbits)

- $\therefore$  {
- Hamiltonian orbits are not isolated from orbit cylinders.
  - energy conservation